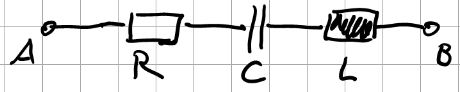
$R = 10\,\Omega \qquad C = 200\,\mu F \qquad L = 3\,mH \qquad f = 50\,Hz$

Reihenschaltung: $\underline{Z}_{ges} = \underline{Z}_{AB}$



$$\underline{Z}_{AB} = \underline{Z}_R + \underline{Z}_C + \underline{Z}_L$$

$$= R + \frac{1}{j\omega C} + j\omega L$$

$$\frac{1}{j} = -j$$

$$= R - j\frac{1}{\omega C} + j\omega L$$

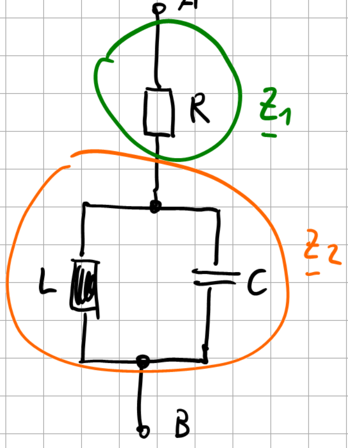
$$= R + j\left(-\frac{1}{\omega C} + \omega L\right)$$

$$= R + j\left(-\frac{1}{2\pi\, 50\,\frac{1}{s} \cdot 200\,\mu F} + 2\pi\, 50\,\frac{1}{s} \cdot 3\,mH\right)$$

$$= 10\,\Omega + j\left(-15{,}91\,\frac{1}{\frac{1}{\cancel{s}}\,\frac{A\cancel{s}}{V}} + 0{,}94\,\frac{V\cancel{s}}{A\cancel{s}}\right)$$

$$\underline{Z}_{AB} = 10\,\Omega - j\,14{,}97\,\Omega$$

Beispiel



$$\underline{Z}_{AB} = R + (\underline{Z}_L \parallel \underline{Z}_C)$$

$$\underline{Z}_{AB} = \underline{Z}_1 + \underline{Z}_2$$

$$\underline{Z}_{AB} = R + \frac{\underline{Z}_L \cdot \underline{Z}_C}{\underline{Z}_L + \underline{Z}_C}$$

$$= R + \frac{j\omega L \cdot \frac{1}{j\omega C}}{j\omega L + \frac{1}{j\omega C}}$$

$$= R + \frac{j\omega L \cdot \frac{1}{j\omega C}}{\frac{j\omega L \cdot j\omega C}{j\omega C} + \frac{1}{j\omega C}}$$

$$= R + \frac{\frac{j\omega L}{\cancel{j\omega C}}}{\frac{j^2\omega^2 LC + 1}{\cancel{j\omega C}}}$$

$$= R + j\,\frac{\omega L}{1 - \omega^2 LC}$$

$$= 10\,\Omega + j\,\frac{2\pi\, 50 \frac{1}{s} \cdot 3\,mH}{1 - \left(2\pi\, 50 \frac{1}{s}\right)^2 3\,mH \cdot 200\,\mu F}$$

$$\underline{Z}_{AB} = 10\,\Omega + j\, 1{,}002\,\Omega$$

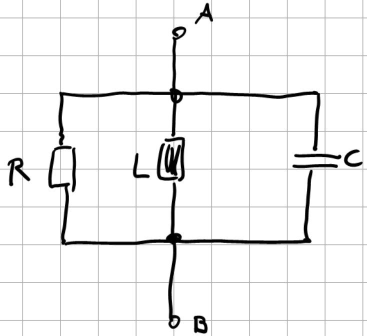
$$\underline{Z}_C = \frac{1}{j\omega C} = -j\,\frac{1}{\omega C}$$

$$\underline{Z}_L = j\omega L \qquad ¿ \quad L^{=}_{Ersatz}$$

$$j\, 1{,}002\,\Omega = j\omega L_{Ersatz}$$

$$1{,}002\,\Omega = \omega L_{Ersatz}$$

$$\Rightarrow L_{Ersatz} = \frac{1{,}002\,\Omega}{2\pi\, 50 \frac{1}{s}} = 3{,}189\, \underbrace{\frac{Vs}{A}}_{H} \cdot 10^{-3}$$



$$\underline{Z}_{AB} = \frac{1}{\underline{Y}_{AB}}$$

$$\underline{Y}_{AB} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$$

$$= \frac{1}{R} + j\left(\frac{-1}{\omega L} + \omega C\right)$$

$$\underline{Y}_{AB} = \frac{1}{10\Omega} + j\left(\frac{-1}{2\pi\, 50\,\frac{1}{s}\cdot 3\,mH} + 2\pi\, 50\,\tfrac{1}{s}\cdot 200\mu F\right)$$

$$= 0{,}1\,\frac{1}{\Omega} + j\left(-1{,}061\,\frac{1}{\Omega} + 0{,}063\,\frac{1}{\Omega}\right)$$

$$= 0{,}1\cdot\frac{1}{\Omega} - j\, 0{,}998\,\frac{1}{\Omega}$$

$$= (0{,}1 - j\, 0{,}998)\,\frac{1}{\Omega}$$

$$\underline{Z}_{AB} = \frac{1}{\underline{Y}_{AB}} = \frac{1}{(0{,}1 - j\,0{,}988)\,\frac{1}{\Omega}}$$

$$= \frac{1}{0{,}1 - j\,0{,}988}\,\Omega$$

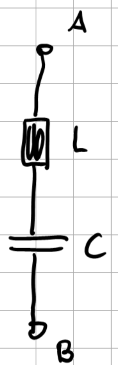
$$= \frac{1}{0{,}1 - j\,0{,}988}\,\underbrace{\frac{0{,}1 + j\,0{,}988}{0{,}1 + j\,0{,}988}}_{\text{konjugiert komplex erweitern}}\,\Omega$$

$$= \frac{0{,}1 + j\,0{,}988}{(0{,}1)^2 + (0{,}988)^2}\,\Omega$$

$$= \frac{0{,}1}{1{,}006}\,\Omega + j\,\frac{0{,}988}{1{,}006}\,\Omega$$

$$= 0{,}1\,\Omega + j\,0{,}98\,\Omega$$

Reihenschaltung aus L und C



$$\underline{Z}_{AB} = \underline{Z}_L + \underline{Z}_C$$

$$= j\omega L + \frac{1}{j\omega C} \qquad \Big|\ \frac{1}{j} = -j$$

$$= j\underbrace{\left(\omega L - \frac{1}{\omega C}\right)}_{\overset{!}{=}0}$$

$$\omega L - \frac{1}{\omega C} = 0$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \sqrt{\frac{1}{LC}}$$

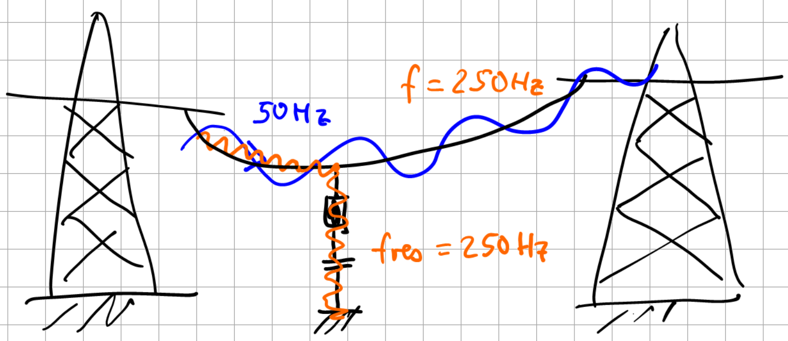
Thomson'sche Schwingungsgleichung

$$\boxed{\omega = \frac{1}{\sqrt{LC}}}$$

Resonanzfrequenz

$$\omega = 2\pi f$$

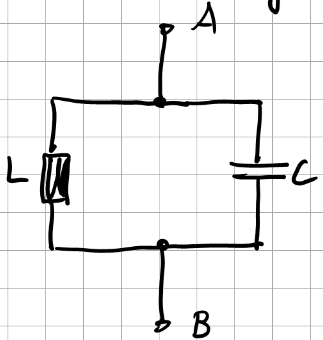
$$f = \frac{1}{2\pi\sqrt{LC}}$$



Bei Resonanzfrequenz verhält sich die L-C-Reihenschaltung wie ein Kurzschluss

Serienresonanz => Kurzschluss

Parallelschwingkreis:



$$\underline{Z}_{AB} = \frac{1}{\underline{Y}_{AB}}$$

$$\underline{Y}_{AB} = \underline{Y}_L + \underline{Y}_C$$

$$= \frac{1}{j\omega L} + j\omega C$$

$$= j\left(-\frac{1}{\omega L} + \omega C\right)$$

$$\underline{Z}_{AB} = \frac{1}{j\left(-\frac{1}{\omega L} + \omega C\right)}$$

$$= j\,\frac{1}{\left(\omega C - \frac{1}{\omega L}\right)} = j\,\frac{1}{\frac{\omega^2 LC - 1}{\omega L}}$$

$$= j\,\frac{\omega L}{\omega^2 LC - 1}$$

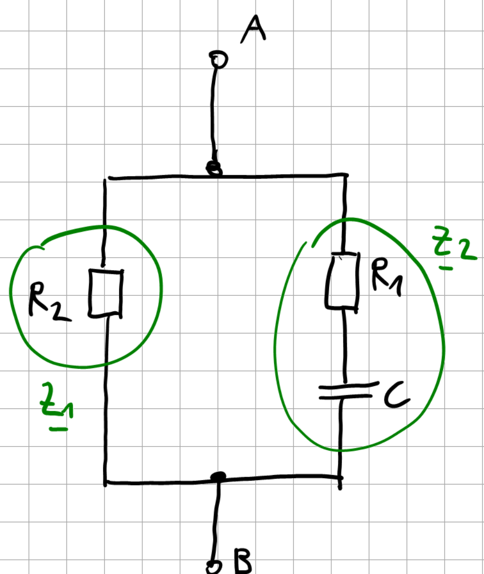
falls $\omega = \omega_{res} = \frac{1}{\sqrt{LC}}$

$$\Rightarrow \underline{Z}_{AB} = j\,\frac{\frac{1}{\sqrt{LC}} \cdot L}{\frac{1}{LC} \cdot LC - 1} = j\,\frac{\sqrt{\frac{L}{C}}}{\text{„}0\text{"}} = j\,\infty$$

Der Parallelschwingkreis L-C verhält sich bei Resonanz wie eine Sperrung (Leerlauf)

Parallelschwingkreis $\Rightarrow$ Leerlauf

Übung:



$\underline{Z}_{AB} = \ldots$

$$\underline{Z}_2 = R_1 + \frac{1}{j\omega C} = \frac{j\omega C R_1 + 1}{j\omega C}$$

$$\underline{Z}_2 = R_2$$

$$\underline{Z}_{AB} = \frac{\underline{Z}_1 \cdot \underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \frac{R_2 \cdot \frac{1+j\omega C R_1}{j\omega C}}{R_2 + \frac{1+j\omega C R_1}{j\omega C}}$$

$$= \frac{R_2 \cdot \frac{1+j\omega C R_1}{\cancel{j\omega C}}}{\frac{j\omega C R_2 + 1 + j\omega C R_1}{\cancel{j\omega C}}}$$

$$\underline{Z}_{AB} = \frac{R_2 \cdot (1+j\omega C R_1)}{1+j\omega C (R_1+R_2)} \qquad \frac{1-j\omega C(R_1+R_2)}{1-j\omega C(R_1+R_2)}$$

$$= \frac{(R_2 + j\omega C R_1 R_2)(1 - j\omega C R_1 - j\omega C R_2)}{1^2 + (\omega C (R_1+R_2))^2}$$

$$= \frac{R_2 + \cancel{j\omega C R_1 R_2} - \cancel{j\omega C R_1 R_2} + \omega^2 C^2 R_1^2 R_2 - j\omega C R_2^2 + \omega^2 C^2 R_1 R_2^2}{N}$$

$$= \frac{R_2 + \omega^2 C^2 R_1 R_2 (R_1+R_2) - j\omega C R_2^2}{N}$$

$$= R_2 \left(\frac{1 + \omega^2 C^2 R_1 (R_1+R_2)}{1+(\omega C(R_1+R_2))^2} - j \frac{\omega C R_2}{1+(\omega C (R_1+R_2))^2} \right)$$